YEAR 10 SCIENCE (GENERAL)
CHEMICAL SCIENCE
REVISION - ATOMIC STRUCTURE, FORMULAE, EQUATIONS

1. The periodic table is useful for chemists as it arranges the elements in a particular way.

(a) What is a group?

Column of the table.

(b) Why are elements put into a particular group?

They have the same chemical characteristics.

(c) What are the group numbers of:

(i) alkali metals? I

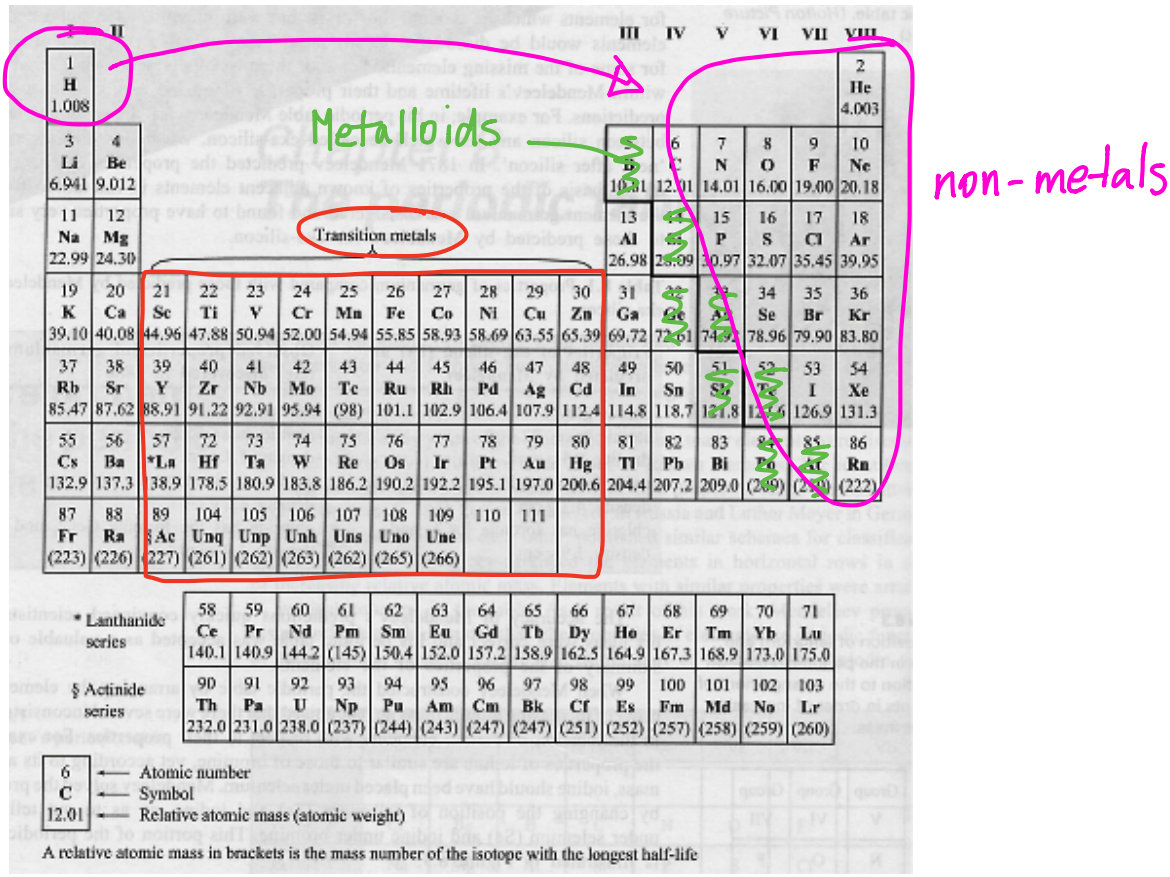
(ii) inert (noble) gases? VIII

(iii) halogens? VII

(d) Where do you find the following elements in the table? Indicate them on the periodic table below, either by labelling or colouring.

(i) transition elements
(ii) metalloids / semi-metals

(e) Where do you find the non-metal elements? Show this on the diagram below.



2. Give a definition or description for the following terms.

(a) atom smallest particle of ordinary matter.

(b) element contains 1 type of atom.

(c) molecule more than 1 atom combined.

(d) compound 2 or more elements chemically combined.

(e) period row of the periodic table.

(f) atomic number number of protons

(g) atomic mass number of protons + neutrons.

(h) ion charged atom due to the gain or loss of electrons.

(i) valence electron electron in the outer shell of an atom.

(g) valency charge on an ion.

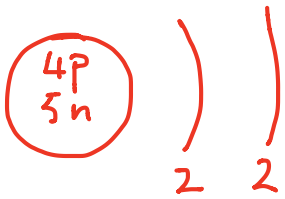
3. Complete the following table.

ELEMENT / ION	PROTON NUMBER	NEUTRON NUMBER	ELECTRON NUMBER
$^{24}_{12}Mg$	12	12	12
$(^{35}_{17}Cl)^{1-}$	17	18	18
$^{56}_{26}Fe$	26	30	26
$(^{27}_{13}Al)^{3+}$	13	14	10
$^{16}_{8}O$	8	8	8

4. Draw clear diagrams to show the electron configurations of the following elements.

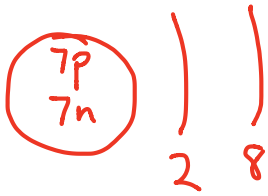
(a) $^{16}_{8}O$

(b) $^{9}_{4}Be$



(c) $(^{39}_{19}K)^{1+}$

(d) $(^{14}_{7}N)^{3-}$



5. Using $(^{35}_{17}Cl)^{1-}$ as an example, explain the following.

(a) Were electrons gained or lost to form the Cl^{1-} ion?

gained

(b) How many electrons were gained or lost?

1 electron gained

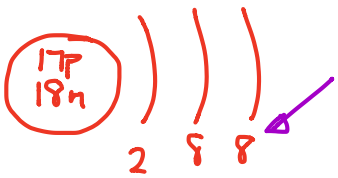
(c) In which group is Cl?

VII

(d) What valency do all members of this group possess?

1+

(e) Explain carefully why the $(^{35}_{17}Cl)^{1-}$ ion is stable, referring to the Octet Rule.
(HINT: Some mention of the noble or inert gases would be appropriate!)



1 extra electron fills the outer shell.

- Now has the same electron configuration as the nearest inert gas (Ar).
- Ion is very stable - has a full valence shell.

6. Complete the following table.

NAME	FORMULA
carbon dioxide	CO_2
aluminium sulphate	$Al_2(SO_4)_3$
dinitrogen tetroxide	N_2O_4
calcium hydroxide	$Ca(OH)_2$
ammonium carbonate	$(NH_4)_2CO_3$
copper (II) chloride	$CuCl_2$
nitrogen dioxide	NO_2
potassium nitrate	KNO_3
zinc sulphide	ZnS
sodium phosphate	Na_3PO_4

7. Balance the following equations.

(a) $2\ PbO + C \rightarrow 2\ Pb + CO_2$

(b) $2\ Fe + 6\ HCl \rightarrow 2\ FeCl_3 + 3\ H_2$

(c) $4\ Al + 3\ O_2 \rightarrow 2\ Al_2O_3$

(d) $C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$

(e) $2\ NH_3 + 3\ CuO \rightarrow 3\ Cu + 3\ H_2O + N_2$

8. Write balanced equations for any of the following mixtures where a precipitate is produced. Be sure to indicate which substance forms a solid - use a subscript $_{(s)}$. If a reaction does not occur, indicate by writing "no reaction".

(a) Calcium chloride solution + sodium carbonate solution.

$CaCl_2(aq) + Na_2CO_3(aq) \rightarrow CaCO_3(s) + 2NaCl(aq)$

(b) Barium hydroxide solution + dilute sulphuric acid.

$Ba(OH)_2(aq) + H_2SO_4(aq) \rightarrow BaSO_4(s) + 2H_2O(l)$

(c) Sodium nitrate solution + sodium chloride solution.

No reaction

(d) Potassium nitrate solution + lead(II) acetate solution.

No reaction

(e) Sodium hydroxide solution + calcium chloride solution.

$2NaOH_{(aq)} + CaCl_{2(aq)} \rightarrow Ca(OH)_{2(s)} + 2NaCl_{(aq)}$

(f) Ammonium carbonate solution + aluminium nitrate solution.

$3(NH_4)_2CO_{3(aq)} + 2Al(NO_3)_{3(aq)} \rightarrow Al_2(CO_3)_{3(s)} + 6NH_4NO_{3(aq)}$

(g) Magnesium nitrate solution + sodium carbonate solution.

$Mg(NO_3)_{2(aq)} + Na_2CO_{3(aq)} \rightarrow MgCO_{3(s)} + 2NaNO_{3(aq)}$

9. The rate of chemical reactions can be affected by several factors. Use your knowledge of these factors to explain the following.

(a) Small pieces of wood are used to start a campfire, rather than large logs.

- Small pieces increases the surface area exposed to air.
- More O_2 can combine with the wood to burn it.

(b) An iron nail slowly glows red-hot in a Bunsen flame, but steel wool burns quickly.

- Nail - limited O_2 in contact with the Fe.
- Wool - lots of O_2 (more surface area).

(c) You can use 1M HCl to produce H_2 gas by reacting it with Mg metal, but 2 M HCl produces more gas in a shorter time.

- 2M HCl has 2x the amount of HCl present.
- Reaction proceeds 2x faster.

(d) Amylase is a digestive enzyme (catalyst) needed by humans to digest carbohydrates.

- Amylase provides a surface for the reaction to occur.
- Less energy is needed for the reaction - goes faster.